# PIA SWARUP

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## EDUCATION

**Olin College of Engineering, Needham, MA** | B.S. in Mechanical Engineering | GPA 3.94/4 Aug 2025 - May 2029

Coursework: Computational Robotics, Mechanical Prototyping, Modeling & Simulation, Instrumentation & Measurement

## EXPERIENCE

**ReviMo** | *Hardware Robotics Intern* | Boston, MA, USA May 2026 - Aug 2026

- Advanced the Niko assistive robot toward production in **Autodesk Fusion**, redesigning 16 parts for strength, **manufacturability**, and **3D printing**; cut part mass by up to 36% and print time by 40% across the redesigned components
- Built automated stress and fatigue **test rigs** with custom control and data-logging electronics, cutting testing time 70%; caught a high-speed TPU part deforming at ~900 cycles and redesigned it to last thousands of cycles
- Bench-tested 5 motors and actuators on torque, efficiency, noise, and heat to select the best fit for each application
- Validated production processes by tuning 3D print settings, materials, and **casting molds**, cutting production time 85%

**Movu NV** | *Warehouse Robotics Intern* | Lokeren, Belgium Feb 2024 - July 2025

- Built custom 3D simulation environments in **Emulate3D** modeling AGV fleets navigating a conveyor system around the Eligo robotic picking arm for warehouse item sorting; used the results to shape the conveyor layout around Eligo
- Redesigned AGV traffic control logic in a real-world-calibrated **Emulate3D** model, raising simulated throughput 10%

**Olin Baja SAE** | *Project Manager* | Needham, MA, USA Sep 2025 - Present

- Lead a 50+ member team through full vehicle design and build; secured 9+ sponsors; ran Baja SAE Ohio 2026 logistics
- Designed and **FEA**-validated wheel hubs under bump, jump, and tree-impact load cases; reduced mass ~50% while maintaining a factor of safety of ~3, with a focus on **Design for Manufacturing**
- Redesigned the pedal box as a removable, top-mounted steel structure, iterating in **FEA** under ~1,000 N to cut weight 45%
- Sized the front brakes and selected COTS components based on safety factor, braking force, cost, and fit within the rim
- Modeled CVT airflow to predict internal temperatures and optimize guard geometry for a 90 °C operating target

## PROJECTS

**Autonomous Self-Balancing Bicycle** | *Lead Mechanical Engineer* Feb 2026 - May 2026

- Led mechanical design and manufacturing of a self-balancing bicycle; **waterjet**-cut and **milled** a lightweighted steel reaction wheel, machined an aluminum motor interface, and 3D printed a servo-based steering system and mounts
- Integrated the mechanical system with an **STM32**-based **PID** controller to achieve closed-loop dynamic balance
- Co-authored a **technical paper** on the modular **open-source** platform, published with full CAD, firmware, and a sub-$1,000 bill of materials on GitHub

**Soil Sampling Drone Attachment** | *Mechanical & Hardware Engineer* Jan 2026 - May 2026

- Designed a **patent-pending** retractable slung-load claw anchor and auger sampler for an autonomous farm drone fleet
- Met the 680 g payload limit and reached $R^2$ = 0.94 sampling accuracy; won Best Use of Drones at Farm Robotics Challenge

**Soil Probing Ground Rover** | *Lead Mechanical Engineer* Sep 2026 - Present

- Designed a retractable gantry for the farm-ng Amiga rover to autonomously probe soil moisture and health in farm fields

**Autonomous Navigation & Computer Vision** | *Robotics Software Engineer* Sep 2026 - Present

- Built a **ROS2** state machine fusing odometry, lidar, and bump data so a vacuum robot follows people and drives a pentagon
- Coded a likelihood-field particle filter using augmented Monte Carlo localization to solve the kidnapped-robot problem; tested with recorded ROS bags in **Rviz2** and in **Gazebo** simulation; converged in 7 update cycles
- Building a computer-vision eye-tracking system for hands-free cursor control and eye-strain break reminders

## SKILLS

**Design & Fabrication**: SolidWorks, Onshape, Fusion, FEA, DFM/DFA, CNC & Manual Machining, Waterjet, Welding, 3D Printing
**Robotics & Controls**: Python, ROS2, Rviz2, Gazebo, MATLAB/Simulink, Emulate3D, PID Control, STM32, Arduino, Raspberry Pi
**Languages**: English (Native), German (Native), Dutch (Native), French (Basic)